

Algebra: Rational Exponents & Radical Expressions

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DIRECTIONS

Convert between radical and rational exponent forms. Evaluate numeric expressions.

1 Evaluate:

$$4^{5/2}$$

Answer: _____

2 Evaluate:

$$32^{3/5}$$

Answer: _____

3 Rewrite using radicals:

$$x^{3/4}$$

Answer: _____

4 Rewrite using radicals:

$$x^{4/3}$$

Answer: _____

5 Evaluate:

$$16^{3/4}$$

Answer: _____

6 Rewrite with rational exponents:

$$\sqrt{x^5}$$

Answer: _____

7 Rewrite using radicals:

$$x^{5/2}$$

Answer: _____

8 Rewrite with rational exponents:

$$\sqrt{x^3}$$

Answer: _____

9 Rewrite with rational exponents:

$$\sqrt[4]{x^7}$$

Answer: _____

10 Evaluate:

$$25^{3/2}$$

Answer: _____



Rational Exponents and Radicals Worksheet

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Answer Key & Solutions

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TEACHER NOTES $x^{(m/n)} = (n\text{th root of } x)^m$. Evaluate by taking the root first to keep numbers small.

1 Evaluate:

$$= 32$$

$$4^{5/2}$$

2 Evaluate:

$$= 8$$

$$32^{3/5}$$

3 Rewrite using radicals:

$$= \sqrt[4]{x^3}$$

$$x^{3/4}$$

4 Rewrite using radicals:

$$= \sqrt[3]{x^4}$$

$$x^{4/3}$$

5 Evaluate:

$$= 8$$

$$16^{3/4}$$

6 Rewrite with rational exponents:

$$= x^{5/2}$$

$$\sqrt{x^5}$$

7 Rewrite using radicals:

$$= \sqrt{x^5}$$

$$x^{5/2}$$

8 Rewrite with rational exponents:

$$= x^{3/2}$$

$$\sqrt{x^3}$$

9 Rewrite with rational exponents:

$$= x^{7/4}$$

$$\sqrt[4]{x^7}$$

10 Evaluate:

$$= 125$$

$$25^{3/2}$$



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